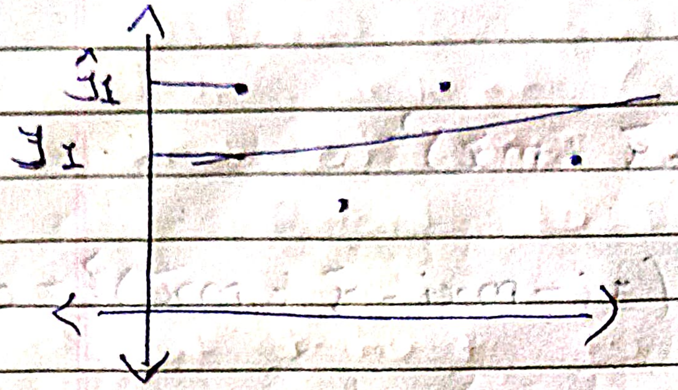
① MAE
(Mean absolute Error)



$$MAE = \frac{|y_1 - \hat{y}_1| + |y_2 - \hat{y}_2| \ldots + |y_n - \hat{y}_n|}{n}$$

$$\boxed{MAE = \frac{\sum_{i=1}^{n} |y_i - \hat{y}_i|}{n}}$$

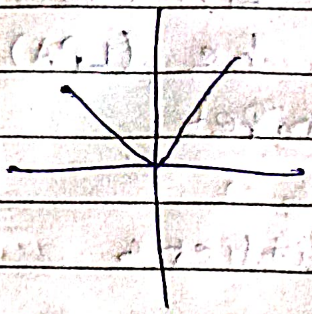
Advan - 1* MAE will be in the units of y

Ex' - if y represents salary in € (euro)

Then if MAE = 1.5 represents 1.5 euro (€)

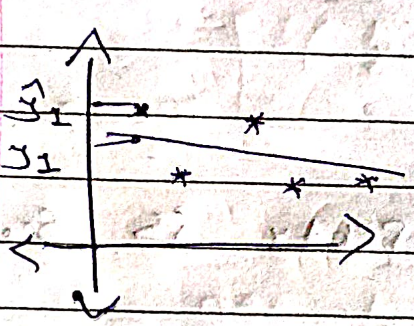
2 - Robust to outlier

* Disadvantage



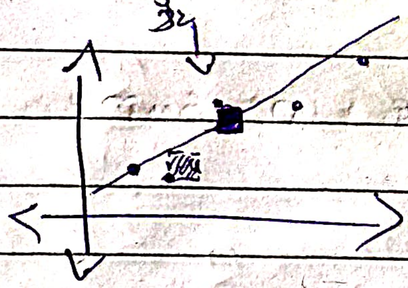
<- graph of modulus is ot differenticable at 0.

② MSE (mean squared error)



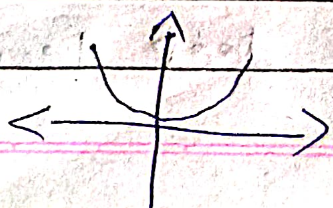
$$MSE = \frac{\sum_{i=1}^{n} (y_i - \hat{y})^2}{n}$$

* what it represent visually



* Adv.

-> can be used as loss function because it is differenciable



* dis

-> if the unit of Y is LPA (Salary) than unit of MSE will be $(LPA)^2$.

-> Hard to interpret

-> not robust to outliers

(3) RMSE

(root mean squared error)

$RMSE = \sqrt{MSE}$

-> most of the Property is same as MSE

* Adv

-> Output will be in same unit as Y.

* dis

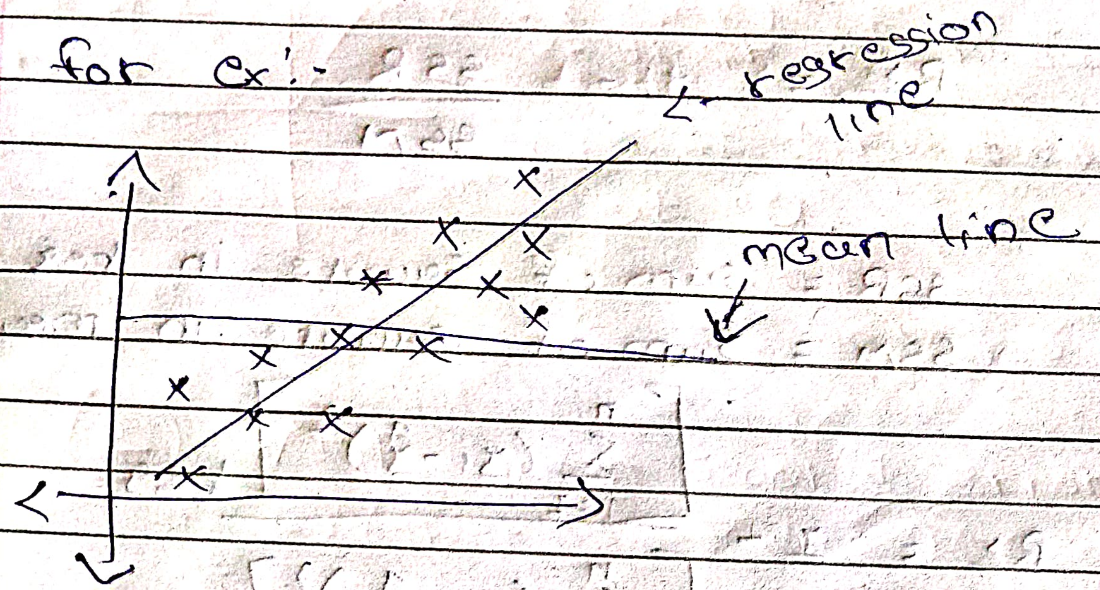
-> Not so robust in outliers

[More used in deep learning]

(4) R2 Score

-> it helps to identify How much one regression line is better than another

for ex:-



-> R2 Score helps to calculate How much your regression line is better than mean line in this ex.

-> R2 Score also known as

1. * Coefficient of Determination (R^2)

$$R^2 = 1 - \frac{SSR}{SST}$$

SSR <- Sum of Squared Regression

SST <- Sum of Squared Total

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - \bar{y})^2}{\sum_i (y_i - \bar{y})^2}$$

2. goodness of fit

* formula can be

$$R^2 = 1 - \frac{SSR}{SSM}$$

SSR = Sum of Square in reg line
SSM = Sum of square in mean line

$$R^2 = 1 - \frac{\left[\sum_{i=1}^{n} (y_i - \hat{y})^2\right]_{reg}}{\left[\sum_{i=1}^{n} (y_i - \bar{y}_i)^2\right]_{mean}}$$

-> Score will be between 0 and 1

1 for best and 0 for worse

-> it can be Negative also

(5) Adjusted R2 Score

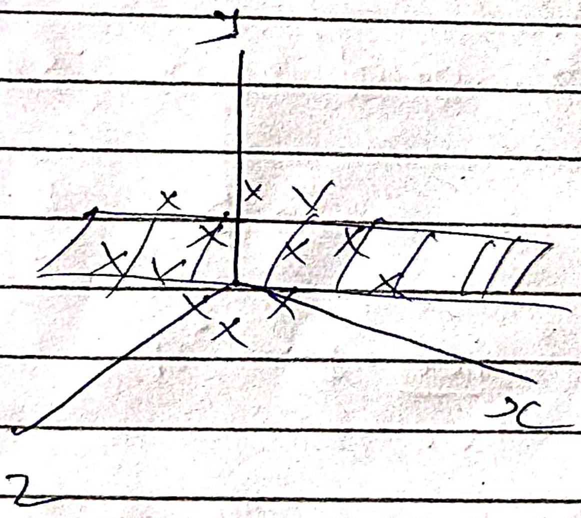
* formula

$$R^2_{adj} = 1 - \left[\frac{(1-R^2)(n-1)}{(n-1-k)} \right]$$

R2 -> R2 Score
n -> no. of ~~cols~~ rows
k -> no. of indipendent cols in data

-> Adj R2 score will help when you have irrelevant column in your data.

* Multiple Linear Regression



$y = mx_1 + mx_2 + b$

↓

- for 3 dimension

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

-> for n dimension

$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$ / or $y = \beta_0 + \sum_{i=1}^{n} \beta_i x_i$

* Coefficients (β_1, β_2) are representing weights

β_0 (=) offset